UNIT I & INTRODUCTION

TOPIC: INTRODUCTION TO CLOUD COMPUTING

1. INTRODUCTION TO CLOUD COMPUTING

- Over the last three decades, businesses that use computing resources have learned to face a vast array of buzzwords like grid computing, utility computing, autonomic computing, on-demand computing and so on.
- A new buzzword named cloud computing is presently in state-of-the-art and it is generating all sorts of confusion about what it actually means.
- In history, the term cloud has been used as a metaphor for the Internet.

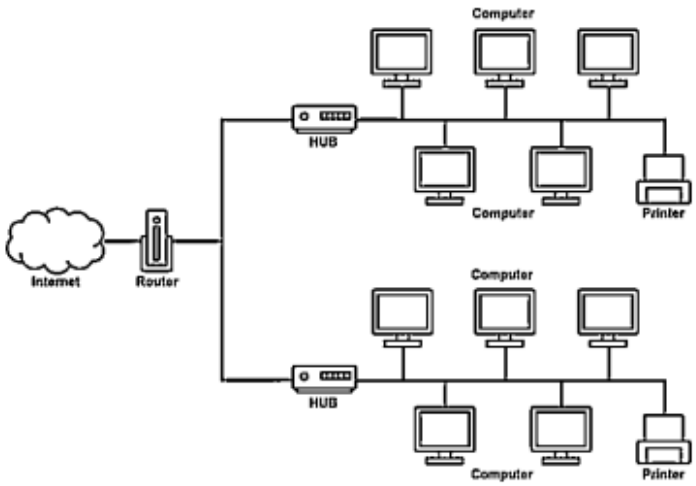

Fig 1.1 Illustration of Network Diagram

- This usage of the term was originally derived from its common illustration in network diagrams as an outline of a cloud and the symbolic representation used to represent the transport of data across the network to an endpoint location on the other side of the network.
- Figure 1.1 illustrates the network diagram which includes the symbolic representation of cloud.

- The cloud computing concepts were initiated in 1961, when Professor John McCarthy suggested that computer time-sharing technology might lead to a future where computing power and specific applications might be sold through a utility-type business model.
- This idea became very popular in the late 1960s, but in mid 1970s the idea vanished away when it became clear that the IT Industries of the day were unable to sustain such a innovative computing model. However, since the turn of the millennium, the concept has been restored.
- Utility computing is the provision of computational resources and storage resources as a metered service, similar to those provided by a traditional public utility company. This is not a new idea.
- This form of computing is growing in popularity, however, as companies have begun to extend the model to a cloud computing paradigm providing virtual servers that IT departments and users can access on demand.
- In early days, enterprises used the utility computing model primarily for non-missioncritical requirements, but that is quickly changing as trust and reliability issues are resolved.
- Research analysts and technology vendors are inclined to define cloud computing very closely, as a new type of utility computing that basically uses virtual servers that have been made available to third parties via the Internet.
- Others aimed to describe the term cloud computing using a very broad, all-inclusive application of the virtual computing platform. They confront that anything beyond the network firewall limit is in the cloud.
- A more softened view of cloud computing considers it the delivery of computational resources from a location other than the one from which the end users are computing.
- The cloud sees no borders and thus has made the world a much smaller place. Similar to that the Internet is also global in scope but respects only established communication paths.
- People from everywhere now have access to other people from anywhere else.
- Globalization of computing assets may be the major contribution the cloud has made to date. For this reason, the cloud is the subject of many complex geopolitical issues.
- Cloud computing is viewed as a resource available as a service for virtual data

centers. Cloud computing and virtual data centers are different one.

- For example, Amazon's S3 is Simple Storage Service. This is a data storage service designed for use across the Internet. It is designed to create web scalable computing easier for developers.
- Another example is Google Apps. This provides online access via a web browser to the most common office and business applications used today. The Google server stores all the software and user data.
- Managed service providers (MSPs) offers one of the oldest form of cloud computing.
- A managed service is an application that is accessible to an organization's IT infrastructure rather than to end users which include virus scanning for email, anti spam services such as Postini, desktop management services offered by CenterBeam or Everdream, and application performance monitoring.
- Grid computing is often confused with cloud computing. Grid computing is a form of distributed computing model that implements a virtual supercomputer made up of a cluster of networked or Inter networked computers involved to perform very large tasks.
- Most of the cloud computing deployments in market today are powered by grid computing implementations and are billed like utilities, but cloud computing paradigm is evolved next step away from the grid utility model.
- The majority of cloud computing infrastructure consists of time tested and highly reliable services built on servers with varying levels of virtualized technologies, which are delivered via large scale data centers operating under various service level agreements that require 99.9999% uptime.

2. Definition of cloud:

- Cloud computing is a model for delivering IT services in which resources are retrieved from the internet through web based tools and applications rather than a direct connection to the server.
- In other words, cloud computing is a distributed computing model over a network and means the ability to run a program on many connected components at a same time.
- In the cloud computing environment, real server machines are replaced by virtual machines.

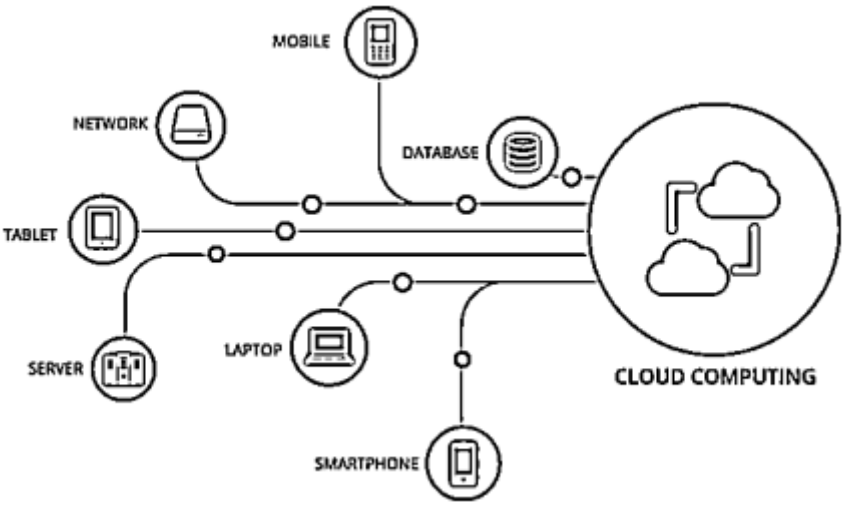

Fig 2.1 Cloud Computing Paradigm

- Such virtual machines do not physically exist and can therefore be moved around and scaled up or down on the fly without affecting the cloud user as like a natural cloud.
- Cloud refers to software, platform, and Infrastructure that are sold as a service. The services accessed remotely through the Internet.
- The cloud users can simply log on to the network without installing anything. They do not pay for hardware and maintenance. But the service providers pay for physical equipment and maintenance.
- The concept of cloud computing becomes much more understandable when one begins to think about what modern IT environments always require scalable capacity or additional capabilities to their infrastructure dynamically, without investing money in the purchase of new infrastructure, all the while without needing to conduct training for new personnel and without the need for licensing new software.
- The cloud model is composed of three components.

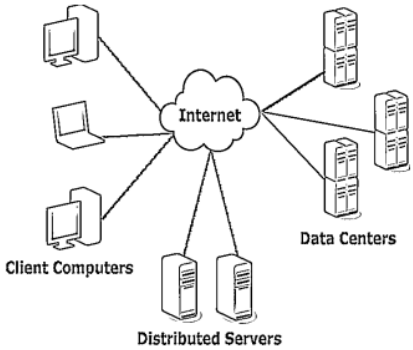

Fig 2.2 Cloud Components

- Clients are simple computers might be laptop, tablet, mobile phone.
- Categories of clients are Mobile clients, Thin clients and Thick clients. ○ Mobile clients which includes smartphones and PDAs.
- Thin clients which include servers without internal hardware. Usage of this type of clients leads to Low hardware cost, Low IT Cost, Less power consumption and less noise.
- Thick clients which includes regular computers.
- Data Center is a collection of servers and it contains clients requested applications.
- Distributed Server in which server is distributed in different geographical locations

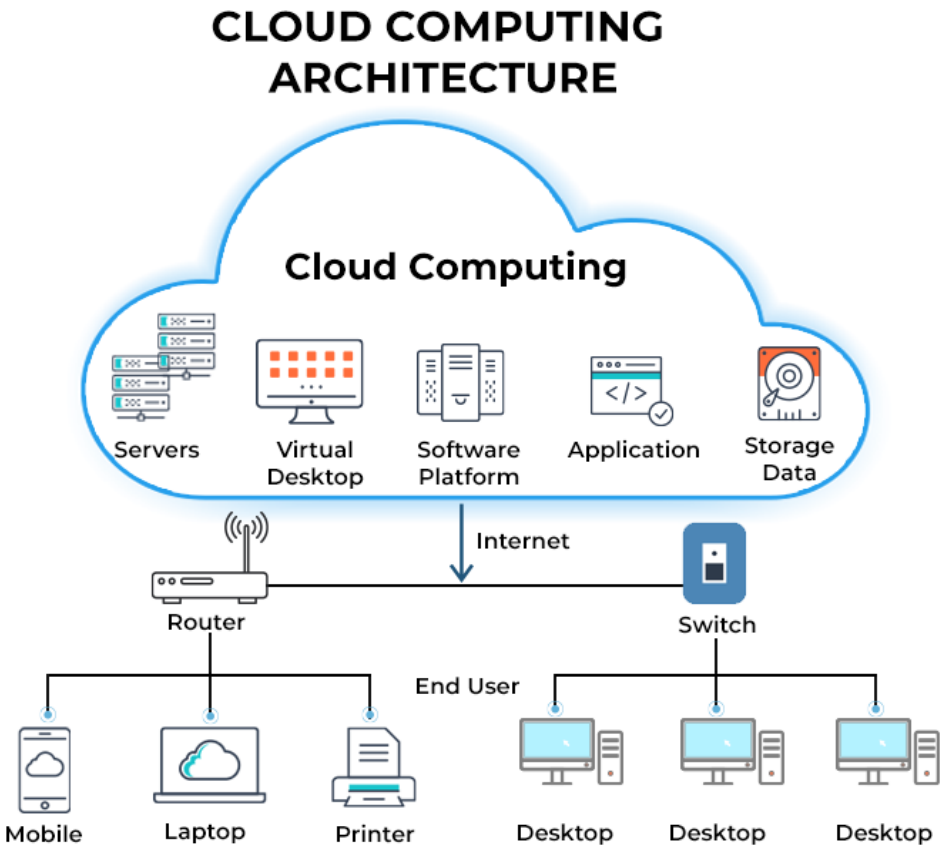

Fig 2.3 Cloud Computing Architecture

3. TYPES OF CLOUD COMPUTING:

i. Public cloud:

- Public cloud are managed by third parties which provide cloud services over the internet to public, these services are available as pay-as-you-go billing mode.
- They offer solutions for minimizing IT infrastructure costs and act as a good option

for handling peak loads on the local infrastructure.

- They are a goto option for small enterprises, which are able to start their businesses without large upfront investments by completely relying on public infrastructure for their IT needs.
- A fundamental characteristic of public clouds is multitenancy.
- A public cloud is meant to serve multiple users, not a single customer. A user requires a virtual computing environment that is separated, and most likely isolated, from other users.

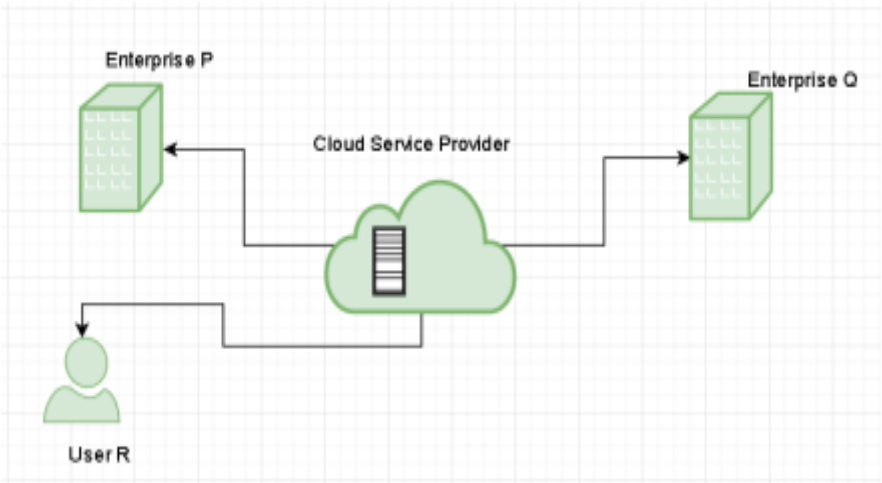

Fig 3.1 Public Cloud

ii. Private cloud:

- Private clouds are distributed systems that work on a private infrastructure and providing the users with dynamic provisioning of computing resources.
- Instead of a pay-as-you-go model as in public clouds, there could be other schemes in that take into account the usage of the cloud and proportionally billing the different departments or sections of an enterprise.

The advantages of using a private cloud are:

1. Customer information protection: In private cloud security concerns are less since customer data and other sensitive information does not flow out of a private infrastructure.

2. Infrastructure ensuring SLAs: Private cloud provides specific operations such as appropriate clustering, data replication, system monitoring and maintenance, and disaster recovery, and other uptime services.

3. Compliance with standard procedures and operations: Specific procedures have to be put in place when deploying and executing applications according to third-party compliance standards.This is not possible in case of public cloud.

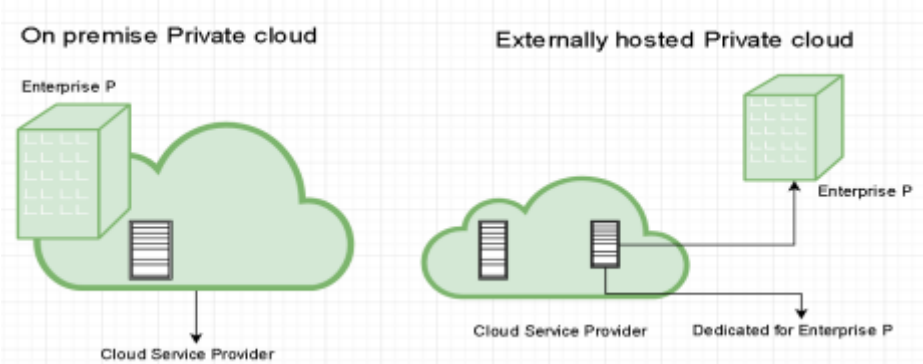

Fig 3.2 Private Cloud

iii. Hybrid cloud:

- Hybrid cloud is a heterogeneous distributed system resulted by combining facilities of public cloud and private cloud.
- For this reason they are also called heterogeneous clouds.
- A major drawback of private deployments is the inability to scale on demand and to efficiently address peak loads.
- Here public clouds are needed. Hence, a hybrid cloud takes advantages of both public and private cloud.

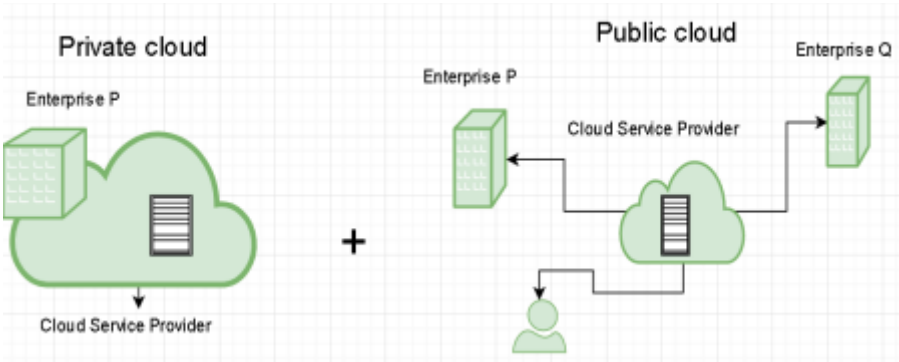

Fig 3.3 Hybrid Cloud

iv. Community cloud:

- Community clouds are distributed systems created by integrating the services of different clouds to address the specific needs of an industry, a community, or a

business sector.

- In community cloud, the infrastructure is shared between organization which have shared concerns or tasks.
- The cloud may be managed by an organization or a third party.

Sectors that use community clouds are:

1. Media industry: Media companies are looking for quick, simple, low-cost way for increasing efficiency of content generation.Most media productions involve an extended ecosystem of partners. In particular, the creation of digital content is the outcome of a collaborative process that includes movement of large data, massive compute-intensive rendering tasks, and complex workflow executions.

2. Healthcare industry: In healthcare industry community clouds are used to share information and knowledge on the global level with sensitive data in the private infrastructure.

3. Energy and core industry: In these sectors, the community cloud is used to cluster set of solution which collectively addresses management, deployment, and orchestration of services and operations.

4. Scientific research: In this organization with common interests of science share large distributed infrastructure for scientific computing.

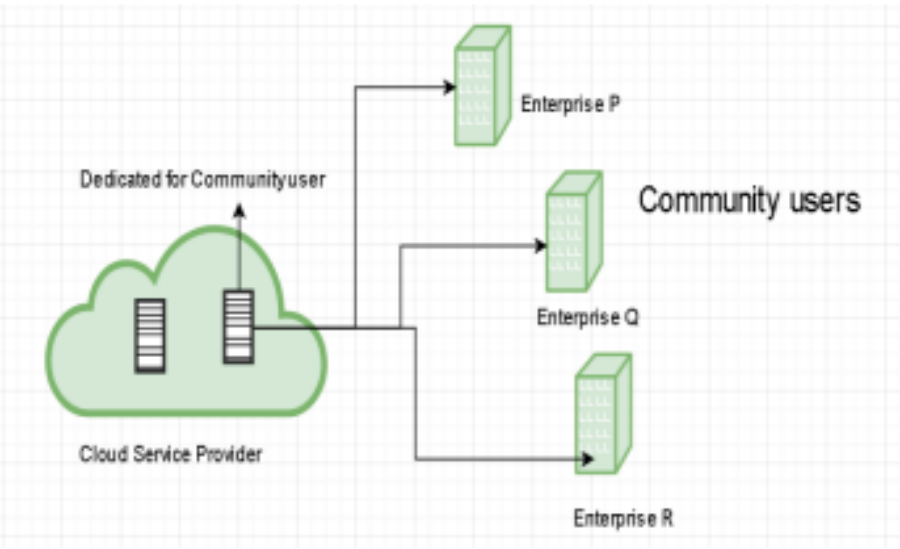

Fig 3.4 Community Cloud

DISCUSSION QUESTIONS:

1. Define Cloud Computing.

2. Illustrate the purpose of the Cloud Computing.

3. List the types of Cloud Computing.

4. Identify the sectors that use community clouds.

5. List the advantages of Private Cloud.